FX Equity Spillover

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1 Introduction

The goal of this model is to analyze and develop a strategy around trading FX using information within equities. There is likely a relationship between a country's equity index and its domestic currency returns. The relationship has been well studied within fixed income markets, specifically interest rate parity, but is less researched within equities. The literature is also not in full agreement. Nevertheless, the relationship can be learned by a generic model and used within a portfolio.[1]

2 Main approach and Theoretical Background

The initial model is developed using FX futures. In this case, the standard front month roll adjusted futures via Bloomberg is suitable. The FX futures are not fully expansive of all FX trading, but their coverage is suitable. FX futures were chosen over spot since they offer better financing, possibly lower transaction cost, account for cost of carry or non-deliverable, have high quality data, and are readily accessible to investors. Of the FX futures that are offered, a subset of the most liquid futures is taken for this model. The initial results of this model were quite strong; therefore, later attempts will expand to less liquid futures, and then later to FX carry. Equity returns are calculated using MSCI ETFs total returns values, which will capture dividend adjustments.

Future	MSCI ETF	FX/Country	Liquidity	Start Date	End Date
AD	EWA	Australian Dollar	Liquid	1996-03	2026-01
BP	EWU	British Pound	Liquid	1996-03	2026-01
BR	EWZ	Brazilian Real	Liquid	2000-07	2026-01
CD	EWC	Canadian Dollar	Liquid	1996-03	2026-01
EC	EZU	Euro Foreign	Liquid	2000-08	2026-01
IS	EIS	Israeli Shekel	Illiquid	2008-03	2026-01
JY	EWJ	Japanese Yen	Liquid	1996-03	2026-01
KO	EWY	Korean Won	Illiquid	2006-09	2026-01
NO	ENOR	Norwegian Krone	Illiquid	2012-01	2026-01
NV	ENZL	New Zealand Dollar	Liquid	2010-09	2026-01
PE	EWW	Mexican Peso	Liquid	1996-03	2026-01
PP	EPOL	Polish Zloty	Illiquid	2010-05	2026-01
SE	EWD	Swedish Krona	Illiquid	2002-05	2026-01
SF	EWL	Swiss Franc	Liquid	1996-03	2026-01
SIR	INDA	Indian Ruppee	Illiquid	2013-01	2026-01
UCA	MCHI	Offshore Chinese Renminbi	Illiquid	2012-09	2026-01

[1] A GitHub repo with the full codebase and results can be found https://github.com/diegodalvarez/FXEquitySpillover

This paper is set up in a similar fashion as [Cenedese et al., 2015]. The initial approach was to build a model fitted to this setup

$$r_{fx,t} = r_{f,t} - r_{d,t} + \epsilon_t \tag{1}$$

For FX Futures returns $r_{fx,t}$, foreign equity $r_{f,t}$, and domestic equity $r_{d,t}$. The FX futures is matched to its respective MSCI ETF. The fitted model via OLS becomes

$$r_{fx,t} = \beta \cdot (r_{f,t} - r_{d,t}) + \alpha + \epsilon_t \tag{2}$$

This is very similar to the no-arbitrage approach that Cenedese finds. For a foreign equity return R_j subject to currency changes $\frac{s_{t+1}}{s_t}$ and a domestic equity return R_h the following conditions are true under no-arbitrage constraints, using a discounting factor m_h.

$$\mathbb{E}\left(R_h \cdot m_h\right) = \mathbb{E}\left(R_j \cdot \frac{s_{t+1}}{s_t} \cdot m_h\right) = 1 \tag{3}$$

Then using covariance rules

$$\mathbb{E}\left(R_h\right) \cdot \mathbb{E}\left(m_h\right) + \operatorname{cov}\left(m_h, R_h\right) = \mathbb{E}\left(R_j\right) \cdot \frac{S_{t+1}}{S_t} \cdot \mathbb{E}\left(m_h\right) + \operatorname{cov}\left(m_h, R_j \cdot \frac{S_{t+1}}{s_t}\right) = 1 \tag{4}$$

Then convert returns via log and take the excess returns. We also assume that $\mathbb{E}(R_h) = \frac{1}{R_f}$ where R_f is the risk-free rate. Then calculate the log excess return

$$\mathbb{E}\left(r_j + \Delta s_{t+1} - r_h\right) = rp_j - rp_h + \eta \tag{5}$$

rp_j and rp_h are the risk premias of the domestic and foreign equities. Then rearrange the formula to get to isolate the FX returns.

$$\mathbb{E}(\Delta s_{t+1}) = -\mathbb{E}(r_j - r_h) + (rp_j - rp_h) + \eta \tag{6}$$

Equation 2 and Equation 6 are very close. The main difference is that equation 2 does not include the risk premia. Cenedese use stochastic discount factors which cannot be directly observed and then use the risk free rate.

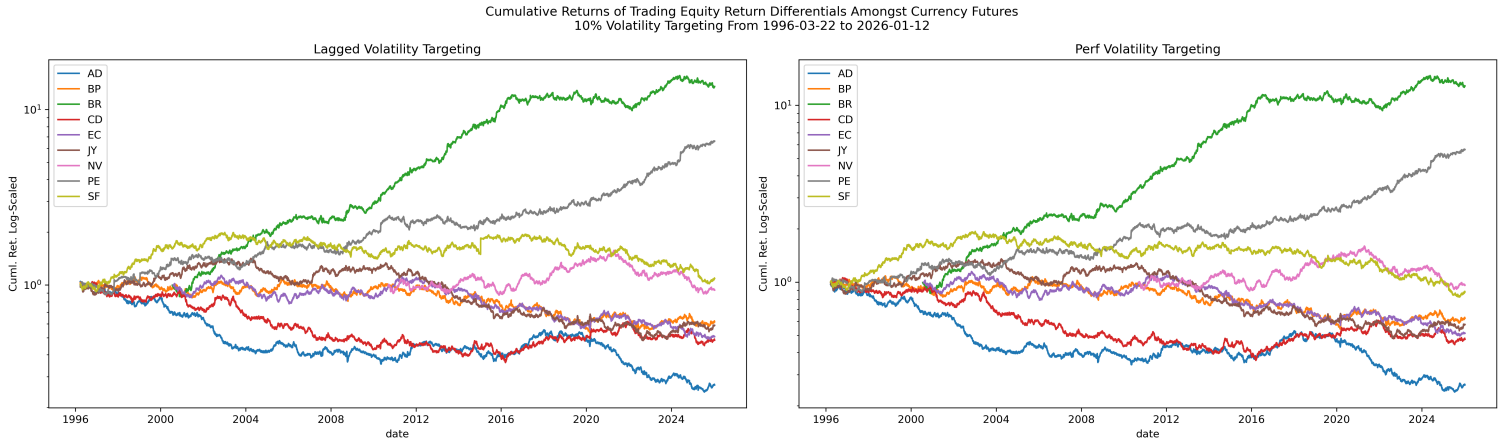
This model differs from Cenedese since it uses the equity returns differential as a factor for FX futures returns. Cenedese calculates cross-sectionalized equity portfolios based on different common equity factors but uses FX returns rather than equities. Therefore, the risk premia component rp_j and rp_h can vary depending on the factor chosen: dividend yield, value, momentum. This model uses the equity returns differential. Moving to Cenedese's approach or incorporating equity factor returns would be a great addition but require modeling the risk premia(s) as well.

There is some existing literature on the relationship between FX returns and equities. The first comes from [Hau and Rey, 2006] who argue that FX returns should offset equity returns through portfolio hedging of international investors, which implies a negative correlation. Other contrasting arguments by [Froot et al., 1992], [Bohn and Tesar, 1996] link FX returns to equity momentum, implying a positive correlation. The original intention of this strategy was attempting to exploit the latter, although the setup of the OLS regression and residual optimization does not infer one way or another.

3 Trading Z-Scores

Start by calculating the 30d exponentially weighted z-score of the difference in daily foreign-domestic equity returns. Then apply that z-score as a signal and apply a 10% annualized volatility target rebalanced daily to each signal's returns. Perfect volatility targeting and lagged volatility targeting are used. The correct approach is lagged volatility targeting, but due to the resolution of the data (daily) at times, large intraday volatility swings can result in large losses. It is fair to assume that an investor (or broker) would adjust positions throughout the day if something extraordinary happens. These types of events are more likely to occur in commodities and not in currencies but the analysis is still useful since it represents the sharpe lost to not implementing intraday volatility targeting.

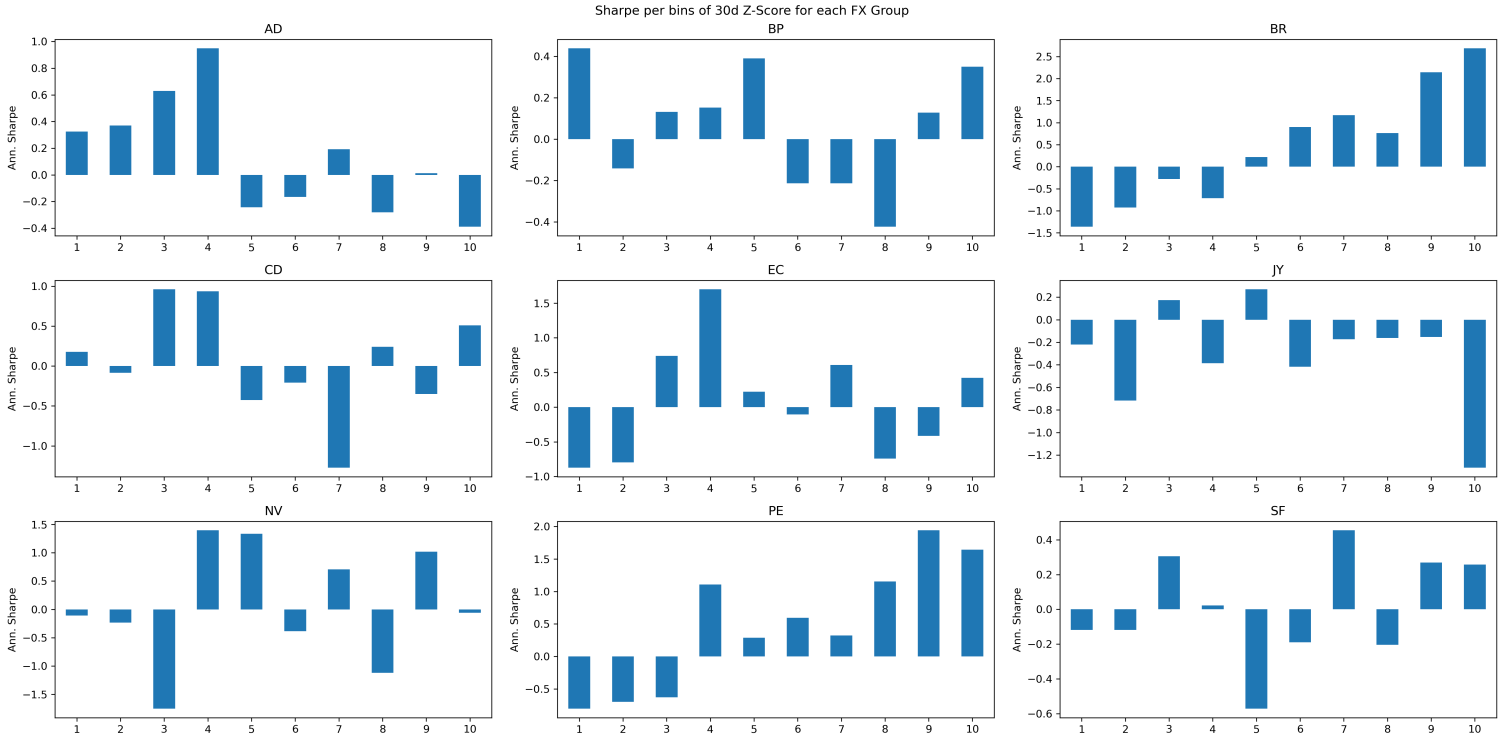
Below are the results of trading the FX futures. Although the results are not strong it appears that the returns are consistent, implying that some fitting-of-data can be used to optimize the signal.



Below is a table of the annualized sharpes.

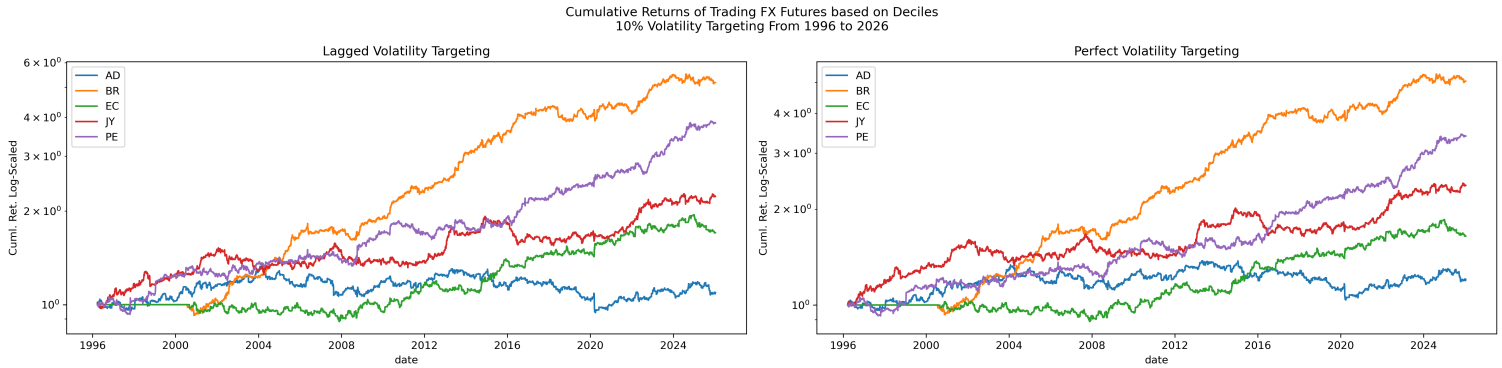
Fut Ticker	AD	BP	BR	CD	EC	JY	NV	PE	SF
Lagged Vol	-0.384	-0.110	1.074	-0.192	-0.219	-0.125	0.006	0.671	0.080
Perfect Vol	-0.413	-0.114	1.091	-0.207	-0.222	-0.142	0.021	0.654	0.005

To optimize this signal calculate the full sample decile of the z-score and then calculate the sharpe per bins. Below is the table for the sharpe. Currencies such as Brazil (BR) and Mexico (PE) show clear signs of momentum-based FX currency returns (positive FX-Equity correlation) while Australia shows the opposite.



To optimize these signal take the outer 2 deciles. Then query out the outer deciles that are not the same sharpe sign, and then trade the outer deciles with respect to the sharpe.

Below are the results of trading the optimized Z-score FX Portfolio. There is a noticeable improvement. The optimization is a quick way to reduce noise, transaction cost, and de-correlate the strategies.



Below is a table of the annualized sharpes over the full period.

Vol Target	AD	BR	EC	JY	PE
Lagged	0.125	1.638	0.572	0.714	1.116
Perfect	0.212	1.697	0.569	0.796	1.118

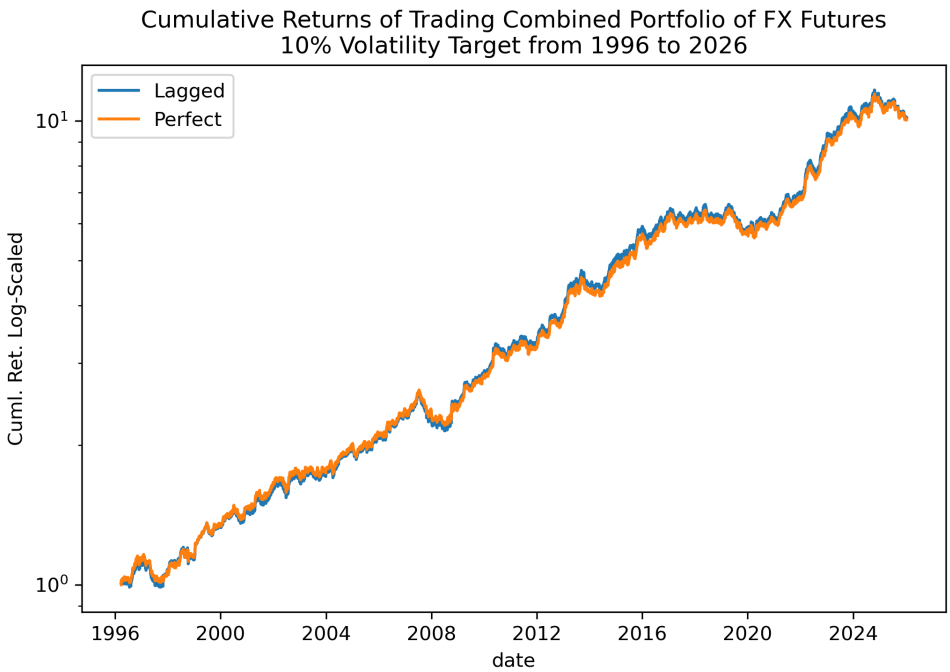
The correlation of the z-score decile optimized strategies are low, below is a correlation matrix.

Vol Target	Lagged					Perfect				
	AD	BR	EC	JY	PE	AD	BR	EC	JY	PE
AD	1.000	-0.187	-0.529	-0.156	-0.173	1.000	-0.192	-0.533	-0.151	-0.185
BR	-0.187	1.000	0.094	-0.016	0.411	-0.192	1.000	0.104	-0.018	0.416
EC	-0.529	0.094	1.000	0.296	0.154	-0.533	0.104	1.000	0.287	0.162
JY	-0.156	-0.016	0.296	1.000	0.073	-0.151	-0.018	0.287	1.000	0.077
PE	-0.173	0.411	0.154	0.073	1.000	-0.185	0.416	0.162	0.077	1.000

The sharpe of the strategies needs to be adjusted for the time when there was no position held. The overall goal of the decile optimization is to residualize the signal and trade the residualization if its optimal. At most the model can only be invested $\frac{2}{5}$ of the time. Therefore, the adjusted sharpe portfolio is

Vol Target		Lagged		Perfect
	Adj Sharpe	Raw Sharpe	Adj Sharpe	Raw Sharpe
Fut				
AD	0.177	0.125	0.300	0.212
BR	1.661	1.638	1.721	1.697
EC	0.813	0.572	0.809	0.569
JY	0.718	0.714	0.801	0.796
PE	1.133	1.116	1.134	1.118

Taking equal weight which will should be roughly equal volatility since each strategy is volatility targeted will result in the following portfolio.



Vol Target	Adj Sharpe	Raw Sharpe
Lagged	1.119	1.119
Perfect	1.184	1.184

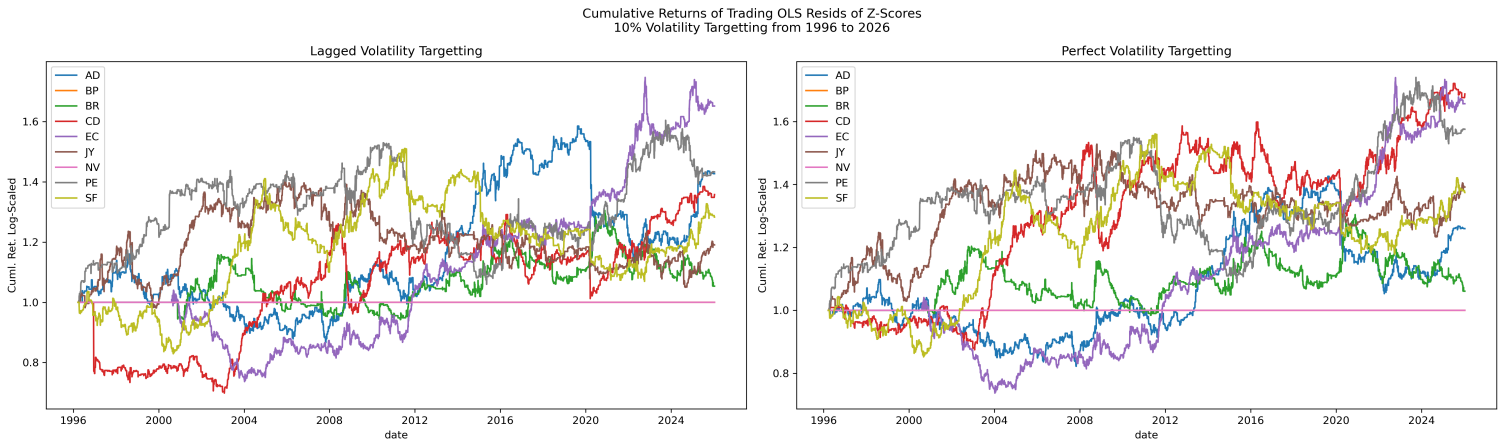
4 Trading OLS Residuals

This section will cover trading the OLS regression from equation 2. Rather than taking the raw returns differential, take the z-score of the differential. Below is a table of that regression, one regression uses the raw returns of FX future and the other fits the z-score to the 10% volatility-targeted FX future's returns.

Ticker	Param Target	Param Val α	Param Val Z-Score	p-value α	p-value Z-Score	t-stat α	t-stat Z-Score
AD	No Target	0.0001	-0.0001	0.4709	0.1808	0.7210	-1.3384
	Vol Target	0.0000	-0.0001	0.7574	0.1130	0.3089	-1.5850
BP	No Target	0.0000	-0.0000	0.7450	0.8745	0.3252	-0.1579
	Vol Target	0.0000	-0.0001	0.5473	0.4500	0.6019	-0.7555
BR	No Target	0.0003	0.0008	0.0204	0.0000	2.3191	5.9876
	Vol Target	0.0001	0.0005	0.0575	0.0000	1.8999	5.7306
CD	No Target	0.0000	-0.0000	0.7744	0.7839	0.2867	-0.2742
	Vol Target	-0.0000	0.0000	0.6989	0.9897	-0.3868	0.0129
EC	No Target	0.0000	0.0000	0.6756	0.5404	0.4185	0.6123
	Vol Target	0.0000	0.0000	0.8577	0.9223	0.1793	0.0975
JY	No Target	-0.0001	-0.0001	0.0807	0.1825	-1.7467	-1.3332
	Vol Target	-0.0002	-0.0001	0.0048	0.1756	-2.8202	-1.3547
NV	No Target	0.0000	0.0001	0.8094	0.4335	0.2412	0.7833
	Vol Target	-0.0000	0.0001	0.8723	0.3538	-0.1608	0.9273
PE	No Target	0.0002	0.0004	0.0142	0.0000	2.4515	4.8033
	Vol Target	0.0002	0.0003	0.0026	0.0000	3.0096	4.3946
SF	No Target	0.0000	0.0001	0.8737	0.4176	0.1590	0.8106
	Vol Target	0.0000	0.0000	0.9672	0.7535	0.0411	0.3140

Its fair to admit that the model does not fit the data well. Most of the parameters have weak p-values and t-stats. More work is needed in this area to get a better fit of the data which may require preprocessing changes or a different model such as logistic. Trading the raw residuals of the OLS model leads to weak sharpe ratios. Applying the same decile optimization and volatility target leads to a slightly better portfolio.

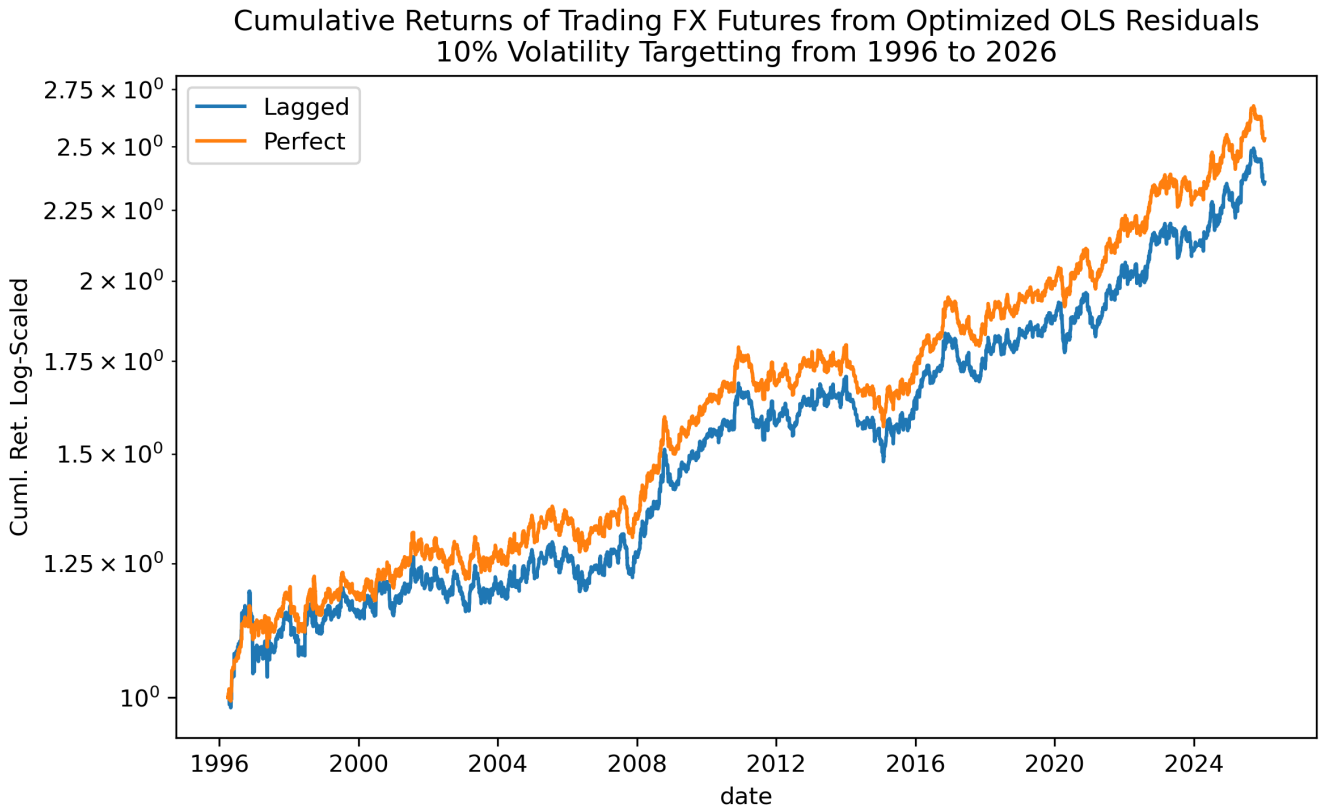
Below are the returns of trading the each optimized signal using the OLS returns



Below is a table of the sharpes of those strategies

Vol. Target	Future	AD	BP	BR	CD	EC	JY	NV	PE	SF
Lagged	Adj Sharpe	0.335	NaN	0.102	0.274	0.542	0.192	NaN	0.339	0.254
	Raw Sharpe	0.220	NaN	0.062	0.180	0.329	0.126	NaN	0.222	0.167
Perfect	Adj Sharpe	0.251	NaN	0.110	0.503	0.559	0.342	NaN	0.449	0.329
	Raw Sharpe	0.165	NaN	0.066	0.331	0.340	0.225	NaN	0.294	0.216

Taking an equal weight which is roughly equal volatility leads to the following portfolio



The sharpes of the portfolio are

group	Adj Sharpe	Raw Rtn
Lagged	0.548	0.502
Perfect	0.650	0.596

References

[Bohn and Tesar, 1996] Bohn, H. and Tesar, L. L. (1996). U.s. equity investment in foreign markets: Portfolio rebalancing or return chasing? *American Economic Review*, 86(1):77–81.

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